

1.

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Hardware or Configuration Issue
1. “CMOS Checksum Error”	Usually indicates a weak/dead CMOS battery or incorrect BIOS settings.
2. “No Boot Device Found”	The HDD/SSD is not detected or no bootable operating system is available.
3. “Memory Error” / “Memory Size Decreased”	Usually points to loose, faulty, or incompatible RAM .
4. “CPU Fan Error”	The CPU cooling fan may be disconnected, blocked, or failing.
5. “Keyboard Error”	The keyboard is disconnected, faulty, or not detected by the system.

2.

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

AMI BIOS: 3 Beeps

For a **traditional AMI BIOS**, **3 short beeps** typically indicate a **base-memory read/write test failure**, meaning the system is having trouble communicating with RAM.

Two troubleshooting steps:

1. **Power off the PC and reseal the RAM:** Remove the RAM sticks and firmly reinstall them in the correct slots. Check for dust or poor contact.
2. **Test the RAM individually:** If there are multiple RAM sticks, test one stick at a time and try a different slot or a known-good compatible RAM module.

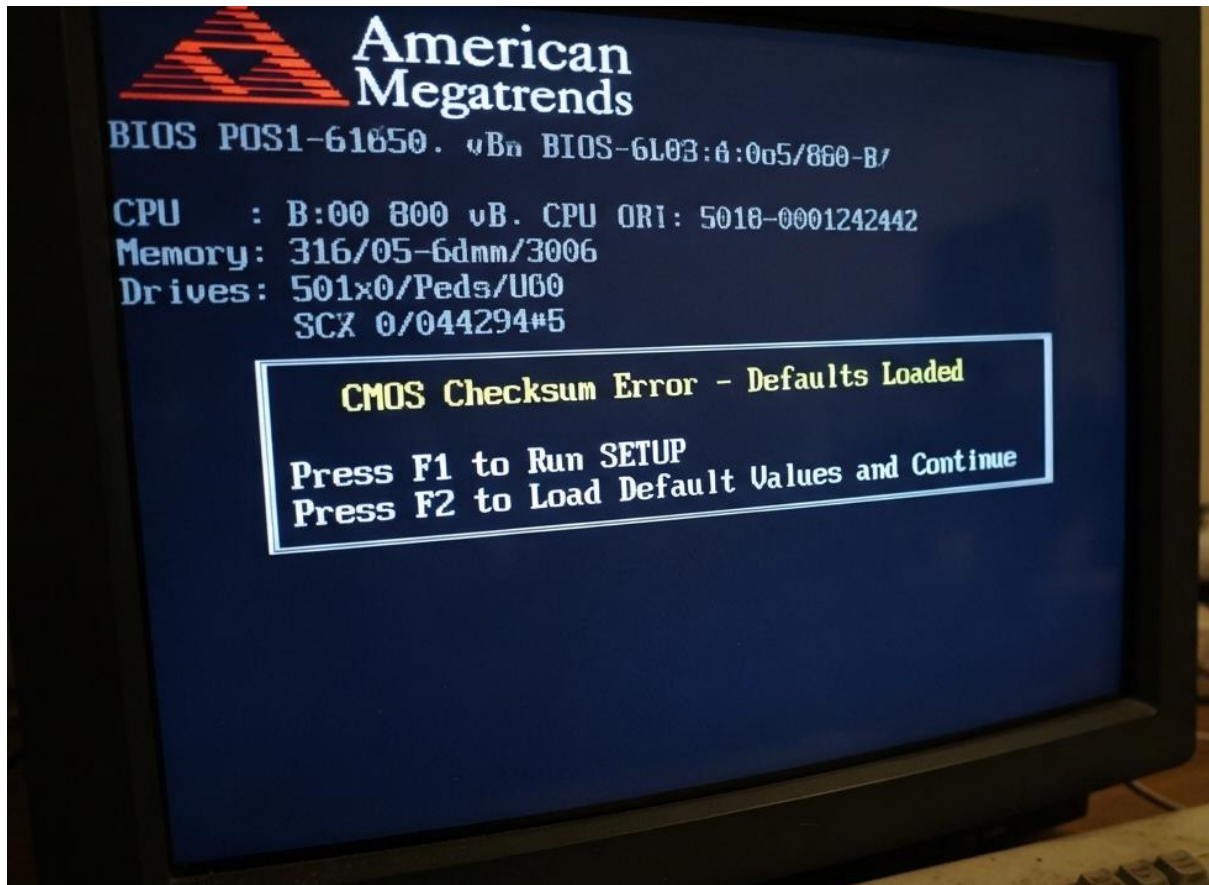
3.

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



4.

You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.



What "Defaults Loaded" Means

The phrase "**Defaults loaded**" in the context of a CMOS checksum error means that the system BIOS detected that the configuration data stored in the CMOS memory (such as boot order, hardware configurations, date, and time) was corrupted or failed its verification check.

To prevent the computer from failing to boot entirely, the motherboard automatically wiped the corrupted data and fell back to the factory **default settings** programmed by the manufacturer.

How to Resolve This Issue in a Real System

To permanently fix this issue, you can follow these steps:

- **Replace the CMOS Battery:** The most frequent cause of this error is a depleted or dead coin-cell battery (CR2032) on the motherboard, which fails to retain power when the PC is turned off. Replace it with a new CR2032 battery.
- **Reconfigure and Save BIOS Settings:** After a reset, enter the BIOS setup utility (by pressing the key prompted on screen, such as F1), update your system date/time and boot priorities, and choose **Save & Exit** (commonly F10) to store the new preferences.
- **Manually Clear the CMOS:** If settings remain glitched, power down the system, unplug it, and either remove the CMOS battery for a few minutes or short the CLR_CMOS / JBAT1 jumper pins to completely flush the NVRAM.
- **Update the BIOS Firmware:** If the error continues to happen repeatedly without battery issues, flash the motherboard to the latest stable BIOS version provided by the manufacturer to fix any underlying bugs.